A Discriminative Tensor Representation Model for Feature Extraction and Classification of Multispectral LiDAR Data

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***Abstract*— Multispectral light detection and ranging (MS-LiDAR) systems open the door to the possibility in the 3-D land cover classification at a finer scale using only point cloud data. This article proposes a model based on the tensor representation for multispectral point cloud classification. The proposed method combines the 3-D local spatial structure of each multispectral point by characterizing the point with a second-order tensor. The first mode of the tensor indicates the spatial location and spectral information of each point (i.e., the row of the second-order tensor) and the second mode denotes the neighborhood geometric and spectral structures (i.e., the column of the second-order tensor). Then we develop a novel tensor manifold discriminant embedding (TMDE) algorithm to extract the geometric–spectral features for multispectral point clouds classification. TMDE solves the mapping matrices of each mode by preserving the intraclass samples' distribution further making it more compact and maximizing the distance of different classes. Finally, the support vector machine classifier with the extracted features as input is used to implement the classification of multispectral point clouds. Experiments are conducted on two real multispectral point cloud data sets. The experimental results demonstrate that the proposed method can achieve significant improvements in classification accuracies in comparison with several state-of-the-art algorithms.**

***Index Terms*— 3-D land cover classification, feature extraction, multispectral point cloud, spatial information, tensor manifold discriminant embedding (TMDE).**

I. Introduction

AIRBORNE laser scanning, also known as light detection and ranging (LiDAR), one of the latest high-resolution airborne remote sensing technologies, is used to collect 3-D point cloud data for a large spatial extent. The point cloud data contain both 3-D coordinates and intensity information of land covers. The geometric structure information can be further obtained by using 3-D coordinates [1]. Fine classification of point clouds is the main requirement when dealing with LiDAR point clouds for specific applications [2], [3]. However, using the traditional monochromatic LiDAR data, the land covers are usually classified into several coarse categories (e.g., ground, buildings, and trees) [4]–[6]. During the last decade, intensity information played a relatively small role in the development of using LiDAR data for land cover classification. The classification of land covers at a finer scale (e.g., grasses, bare soil, and impervious ground) cannot be achieved by using only the geometry of 3-D point clouds. LiDAR data combined with multispectral images can facilitate land cover classification [7]–[9]. The salient advantages of two data sources have been shown in the classification of elevated objects such as buildings and trees. Nevertheless, there are many problems, such as geometric registration and shadow occlusion, the most fatal of which is to degrade the classification of land covers from 3-D into 2-D [10]–[12]. The reason for that is the point clouds must be converted to image rasters to combine multispectral images.

Multispectral light detection and ranging (MS-LiDAR) systems, as a novel active remote sensing technology, offer rich multispectral information for the recorded 3-D point clouds. The data collected by MS-LiDAR are named as multispectral point clouds, independent of solar illumination. As both the geometric and spectral information of land covers are available from a single MS-LiDAR, this technology extends the scope of imaging spectroscopy into stereoscopic spectral sensing. The 3-D classification of land covers at a finer scale can be achieved by using only multispectral point clouds.

Compared to fusing passive spectral images with structural information from monochromatic LiDAR [13], MS-LiDAR has some significant advantages.

1) First and most significantly, the spectrum of each point is independent of any others along the point ray, and spectral mixing within each measured point is minimized due to the relatively small instantaneous field of view (IFOV) of the laser pulse.
2) The spectrum of each return is range resolved because MS-LiDAR can detect multiple objects in a point ray which partially occludes the transmitted beam with a single transmission. This cannot be obtained with the fusion of a passive spectral image and monochromatic LiDAR data. The reason for that is that the pixel of a

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passive sensor is not range resolved which irreversibly mixes the spectra of the constituent partially occluding voxels. Furthermore, the fusion process possibly introduces perspective error.

3) The spectrum is independent of illumination conditions such as shading, an uncertain vertical atmosphere, or secondary scatter effect, due to the active nature of MS-LiDAR. The day or night operational capability of MS-LiDAR solves the incapacity of the traditional passive hyperspectral imaging at night.
4) Finally, there is no need for geometric registration between the two information sources, including image-to-image registration and image-to-point cloud registration.

However, multi/hyper spectral LiDAR is still in development. Hernandez-Marin *et al.* [14] simulated the MS-LiDAR response by using monochromatic LiDAR data based on assumptions about material distribution and spectra. Suomalainen *et al.* [15] presented a measurement system to produce a virtual hyperspectral LiDAR data set by data fusion. Specifically, the measurement system consists of two parts. First, backscattered reflectance spectra were collected by using a spectrometer, a cutting-edge technology, and a white-light supercontinuum laser source. Second, a commercial monochromatic LiDAR system was used for ranging. In recent years, researchers have developed some multi/hyper spectral LiDAR system prototypes, which operate mainly in the laboratory. Woodhouse *et al.* [16] from The University of Edinburgh, Edinburgh, U.K., designed the first MS-LiDAR demonstration optimized for detailed structure and physiology measurements in forest ecosystems. A tunable laser operated at four wavelengths (531, 550, 660, and 780 nm) was used to measure profiles of the normalized difference vegetation index (NDVI) and the photochemical reflectance index (PRI). A research team from Heriot-Watt University, Edinburgh, devised another four-wavelength MS-LiDAR system prototype (531, 570, 670, and 780 nm), which made band selection also based on NDVI and PRI [17]. Wei *et al.* [18] designed a multispectral canopy LiDAR system intended for the remote sensing of vegetation reflection. The system operates with four lasers of different wavelengths (555, 670, 700, and 780 nm) which are chosen according to nitrogen stresses that induce changes in the optical properties and spectral reflectance of rice leaves. Finnish Geodetic Institute (FGI) developed the first full-waveform terrestrial hyperspectral LiDAR prototype [19]. It can collect 3-D point clouds with eight channels backscattered reflectance (542, 606, 672, 707, 740, 775, 878, and 981 nm) for every point.

Multiwavelength LiDAR system, which is the preliminary version of MS-LiDAR, has been developed to enhance the applications of the LiDAR system. Comparing with monochromatic LiDAR, the multiwavelength LiDAR system can acquire more 3-D points and backscattered energy [also named laser return intensity (LRI)] at different wavelengths. The backscattered energy is related to the laser wavelengths, target materials, and target roughnesses. Therefore, it can reflect the physical characteristics of the targets. With the multispectral information from backscattered energy, experimental results in [20] show that multiwavelength LiDAR improves the ability to classify different land covers. Now, several researchers have focused on the land cover classification using MS-LiDAR data. Image-based and point-based classifications are two alternative technologies. Most of the researches used the image-based classification technology to process the MS-LiDAR data. First, intensity and height images were generated from multispectral point clouds. Then, the traditional pixel-based classification or the object-based segmentation and classification techniques were directly used for the MS-LiDAR data classification. With individual pixels as the subject of classification, the Mahalanobis distance [21], maximum likelihood [21], [22], and support vector machine (SVM) classifiers [23] were used to classify land covers. Experimental results in [22] demonstrated the capabilities of MS-LiDAR to classify six land covers with about 90% of overall accuracy (OA). In another way, segmentation technology is first used to extract spatially adaptive objects to be classified. With the extracted objects as the subject of classification, SVM [20], random forest classifier [24], [25], and rule-based classifiers [26] were used to classify the objects into different ground cover classes. The object-based image analysis (OBIA) methodology has achieved over 90% of OA for MS-LiDAR data classification. Rasterizing the 3-D point clouds into 2-D images greatly reduces computational complexity, which is suited for land cover classification of large areas [24]. Nevertheless, the land cover classification is degraded from 3-D to 2-D. As a result, some classes of interest such as power lines and fences are lost.

Several studies used the multispectral points as the subject of classification to achieve 3-D land cover classification. Wichmann *et al.* [27] presented an explorative analysis of multispectral point clouds. It focuses on classifying specific spectral signatures and concludes that multispectral point clouds are suitable for the conventional land cover geometrical classification and mapping procedures. Sun *et al.* [28] used principal component analysis (PCA) to extract the features of point distribution in 3-D space, and then the extracted spatial features along with the LRI at different wavelengths were inputted to SVM to achieve the 3-D classification on a small scene in laboratory. The single-return points and multireturn points were separately considered for classification in [29] and [30]. They used SVM to classify single-return points into ten land cover classes (i.e., building 1, building 2, tree, grass, soil 1, soil 2, asphalt 1, asphalt 2, asphalt 3, and concrete) and achieved an OA of 94.7%. The multireturn points were classified into 11 classes (i.e., the above ten classes plus power line) by using a rule-based classifier and achieved an OA of 79.7%. The authors of studies [30]–[32] compared the image-based and point-based classification technologies on the use of MS-LiDAR data for land cover classification. Compared with image-based methods, point-based methods achieved an OA improvement of 4% in [30], 10% in [31], and 3.8% in [32]. In addition, when multispectral points are the subject of classification, the ground under tree canopies and the very narrow land covers such as power lines and fences can be classified.

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There have been quite a few previous point cloud classification efforts where neighborhood information was leveraged to improve the results on monochromatic LiDAR data. Markov random field (MRF) models can incorporate neighborhood information and be applied to point cloud classification. Anguelov *et al.* [33] used a simplified Markov model, called associative Markov networks (AMNs), to perform a point-wise classification. AMNs enforce the preference that neighboring points have the same label. The pairwise potential function in AMNs is constant for a pair of different class labels. It is too rigorous to allow expressing some natural interactions between objects. Shapovalov *et al.* [34] used a general form of pairwise potentials instead of the constant in AMNs to model typical relations of different classes such as "car is likely to be above ground." Niemeyer *et al.* [35] presented a context-based conditional random field (CRF) classifier for the classification of an airborne LiDAR point cloud. Their method incorporates neighborhood information and learns the specific relations of object classes in a training step. Their experimental results showed that CRFs can provide a high potential for urban scene classification using airborne LiDAR point cloud. Niemeyer *et al.* [36] integrated a random forest classifier into the CRF framework to improve the results further. Niemeyer *et al.* [37] proposed a two-layer CRF to incorporate the spatial and semantic information for airborne LiDAR point clouds classification. Their method iterates and mutually propagates context to improve the classification results. This article is different from the above studies in the sense that how to use neighborhood information. The above studies belong to the family of undirected graphical models. The point cloud data are represented by a graph consisting of nodes and edges. Each node corresponds to a point. The edges link pairs of adjacent nodes and, thus, enable modeling neighborhood relations. This article uses a tensor model to represent the multispectral point cloud data and extracts the geometric–spectral features for multispectral point clouds classification, under the umbrella of multilinear algebra.

MS-LiDAR provides the integrated information of 3-D space and spectra. It describes the targets from different information dimensions and enhances the ability to classify land covers. However, the existing studies just used the height and intensity features directly provided by MS-LiDAR, the simple index features (e.g., vegetation index and water index) and geometric features (e.g., linearity and planarity) to classify land covers. Until now, the feature extraction methods tailored to the characteristics of MS-LiDAR data are still few in number. Tensor is the extension of scalars and vectors for describing the data in mathematics. It can treat high-order data as an entirety and has a strict mathematical foundation [38], [39]. Therefore, tensor representation provides a convenient way to capture the intercorrelation among different dimensions [40], [41]. Lu *et al.* [42] proposed multilinear principal component analysis (MPCA) to deal with tensor objects. Zhang *et al.* [43] proposed tensor discriminative locality alignment (TDLA) for the dimensionality reduction of spectral–spatial features extracted from hyperspectral images. Li and Schonfeld [39] proposed direct general tensor discriminant analysis (DGTDA) for high-order data classification and verified it on the face database. After determining the tensor representation of the multispectral point cloud, the above studies based on the tensor analysis can also be used to process multispectral point clouds.

In this article, a geometric–spectral feature extraction model based on tensor representation is proposed, aiming at improving the ability of 3-D land cover classification by using multispectral point clouds for the first time. This model contains two parts. First, we use the tensor to represent unorganized multispectral point clouds. Each point is represented as a second-order tensor, which is the information source for subsequent feature extraction. Second, under the tensor operation framework, we solve the multilinear mapping matrices by preserving the intraclass samples' distribution and maximizing the distance of different classes. After that, the discriminative geometric–spectral features are extracted by using simple tensor operations. The extracted features are then employed for supervised multispectral point cloud classification. The main contributions of this article are threefold.

1) The tensor representation is used to extract the geometric–spectral features for multispectral point clouds classification for the first time. Compared with the vector-based feature representation, tensor, as a high-order data structure, retains more information for classification.
2) The proposed tensor manifold discriminant embedding (TMDE) algorithm can preserve the local structure and discriminative information to extract the geometric–spectral features, which are more effective for classification. Furthermore, the proposed algorithm can deal with the nonlinear nature of the point cloud distribution by constructing the intraclass and interclass neighborhood graphs, which are conducive to achieve a better classification performance.
3) A generalized feature extraction model is provided for high-order data. In this article, each multispectral point is represented as a second-order tensor. We can also represent it as a third-order tensor for processing. Other data sources, such as multi/hyper spectral image and polarimetric synthetic aperture radar (PolSAR) image, can also carry out feature extraction under the proposed model.

The rest of this article is organized as follows. Section II gives a brief introduction to tensor algebra and the tensor representation of multispectral points. In Section III, the proposed TMDE algorithm is introduced in detail for multispectral point clouds feature extraction. The data sets and experimental methods are introduced in Section IV. Experimental results on two real MS-LiDAR data sets are given in Section V. Section VI concludes this article.

II. Tensor Modeling of MS-LiDAR Point Clouds

A. Tensor Algebra

Tensors, as an extension of scalars and vectors, are multidimensional arrays of numbers that preserve higher order structures of input data [38], [39], [44]. Throughout this article,

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scalars are denoted by lowercase and uppercase italic letters, e.g., i, j, C, L; vectors by bold lowercase letters, e.g., $\mathbf{x}$; matrices by bold uppercase letters, e.g., $\mathbf{X}$, $\mathbf{U}$, $\mathbf{G}$, $\mathbf{Z}$; and tensors by calligraphic letters, e.g., $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$. Their elements are denoted with indices in parentheses, where indices are scalars denoted by lowercase italic letters. The order of a tensor is equal to the mode of the tensor. The order of a tensor $\mathcal{X}_i \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ is M, and the dimensionality of its nth mode is I_n. A few definitions [45], [46] that are relevant to this article are given as follows.

Definition 1 (The n-Mode Product): The n-mode product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ by a matrix $\mathbf{U} \in \mathbb{R}^{P \times I_n}$, denoted by $\mathcal{X} \times_n \mathbf{U}$, is a tensor and its size is $I_1 \times \cdots \times I_{n-1} \times P \times I_{n+1} \times \cdots \times I_M$. The elements of the n-mode product are defined by

$$(\mathcal{X} \times_n \mathbf{U})(i_1, \ldots, i_{n-1}, p_n, i_{n+1}, \ldots, i_M) = \sum_{i_n}^{I_n} \mathcal{X}(i_1, i_2, \ldots, i_M) \cdot \mathbf{U}(p_n, i_n). \quad (1)$$

Definition 2 (The n-Mode Unfolding): The n-mode unfolding of an M-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$, denoted by $\mathcal{X} \Rightarrow_n \mathbf{X}^{(n)}$, is a matrix $\mathbf{X}^{(n)} \in \mathbb{R}^{I_n \times \prod_{i=1, I \neq n}^{I_M} I_i}$. The unfolding matrix $\mathbf{X}^{(n)}$ is the ensemble of vectors in $\mathbb{R}^{I_n}$ obtained by keeping index i_n and varying the other indices. The elements of $\mathbf{X}^{(n)}$ are defined by

$$\mathbf{X}^{(n)}(i_n, k) = X(i_1, i_2, \ldots, i_m) \quad k = 1 + \sum_{j=1, j \neq n}^{M} (i_j - 1) \prod_{q=j+1, q \neq n}^{M} I_q. \quad (2)$$

Definition 3 (Tensor Scalar Product): The scalar product of two tensors $\mathcal{X}, \mathcal{W} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ is a scalar defined by

$$\langle \mathcal{X}, \mathcal{W} \rangle = \sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \cdots \sum_{i_M=1}^{I_M} \mathcal{X}(i_1, i_2, \ldots, i_M) \cdot \mathcal{W}(i_1, i_2, \ldots, i_M). \quad (3)$$

The Frobenius norm of tensor $\mathcal{X}$ is defined by

$$\|\mathcal{X}\|_F = \sqrt{\langle \mathcal{X}, \mathcal{X} \rangle}. \quad (4)$$

The distance between tensors $\mathcal{X}$ and $\mathcal{W}$ can be measured by the Frobenius norm $\text{dist}(\mathcal{X}, \mathcal{W}) = \|\mathcal{X} - \mathcal{W}\|_F$.

B. Tensor Representation of Multispectral Point Clouds

MS-LiDAR utilizes active white light produced by a supercontinuum laser source to survey the (x, y, z) location of object surfaces and measures the return intensity at different wavelengths by using diffraction grating. MS-LiDAR images the environment by unambiguously associating a vector in the spectral space with each vector in the 3-D world space. It produces a 3-D point with multispectral LRI: $(x, y, z, \text{LRI}\lambda)$, where LRI$\lambda$ is the laser return intensity as a function of the wavelength λ. Fig. 1 shows the information structure of a typical MS-LiDAR data; multispectral point clouds generated from the Titan sensor. The geometric and spectral information

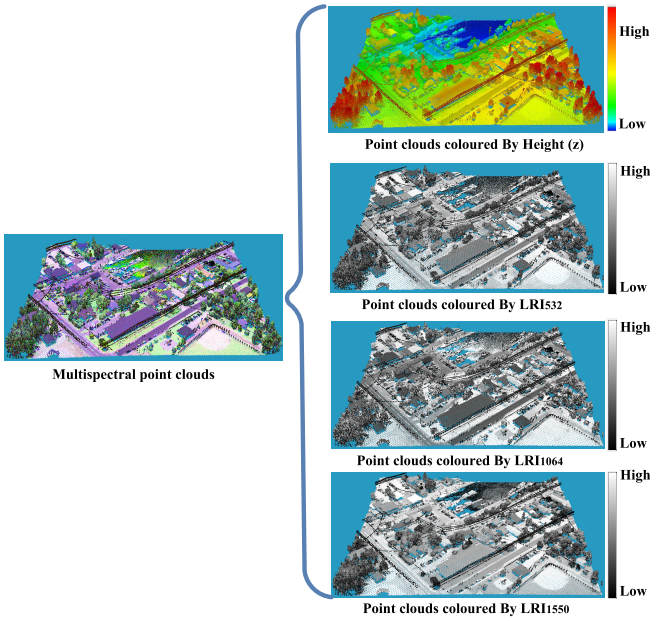


Fig. 1. Information structure of the multispectral point clouds.

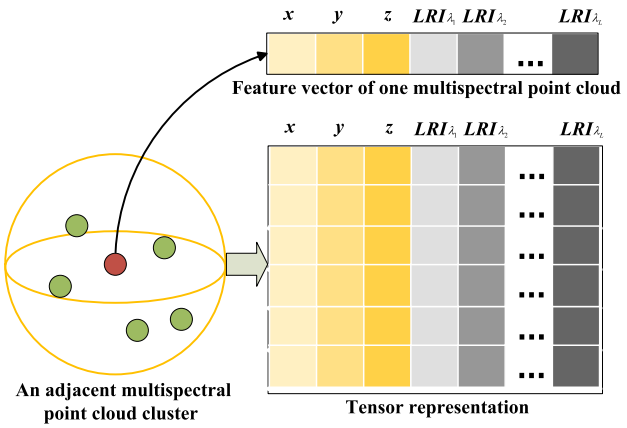


Fig. 2. Vector and tensor representations for a multispectral point cloud.

is available from a single measurement. The single multispectral point provides the spatial location (x, y, z) and spectral (LRIλ) information. The geometric structure information can be obtained from adjacent point clusters.

We represent a multispectral point as a vector $\mathbf{x} = [x, y, z, \text{LRI}_{\lambda_1}, \text{LRI}_{\lambda_2}, \ldots, \text{LRI}_{\lambda_L}]$, where L is the number of bands in the MS-LiDAR data. For each point in the MS-LiDAR data, the k nearest points found by the Euclidean distance are adopted to represent the local geometric and spectral information of the centric point, as shown in Fig. 2. The proposed tensor geometric–spectral feature representation for the analyzed point $\mathbf{x}_c$ is constructed by

$$\mathbf{X}_c^k = \left[\mathbf{x}_c, \mathbf{x}_c^1, \mathbf{x}_c^2, \ldots, \mathbf{x}_c^{k-1}\right] \in \mathbb{R}^{(L+3) \times k} \quad (5)$$

where $\mathbf{x}_c^i$ $(i = 1, 2, \ldots, k-1)$ is the feature vector of one point in the k nearest points.

The proposed tensor representation of a multispectral point has two modes which conduce to better represent the multispectral point cloud's geometric–spectral features.

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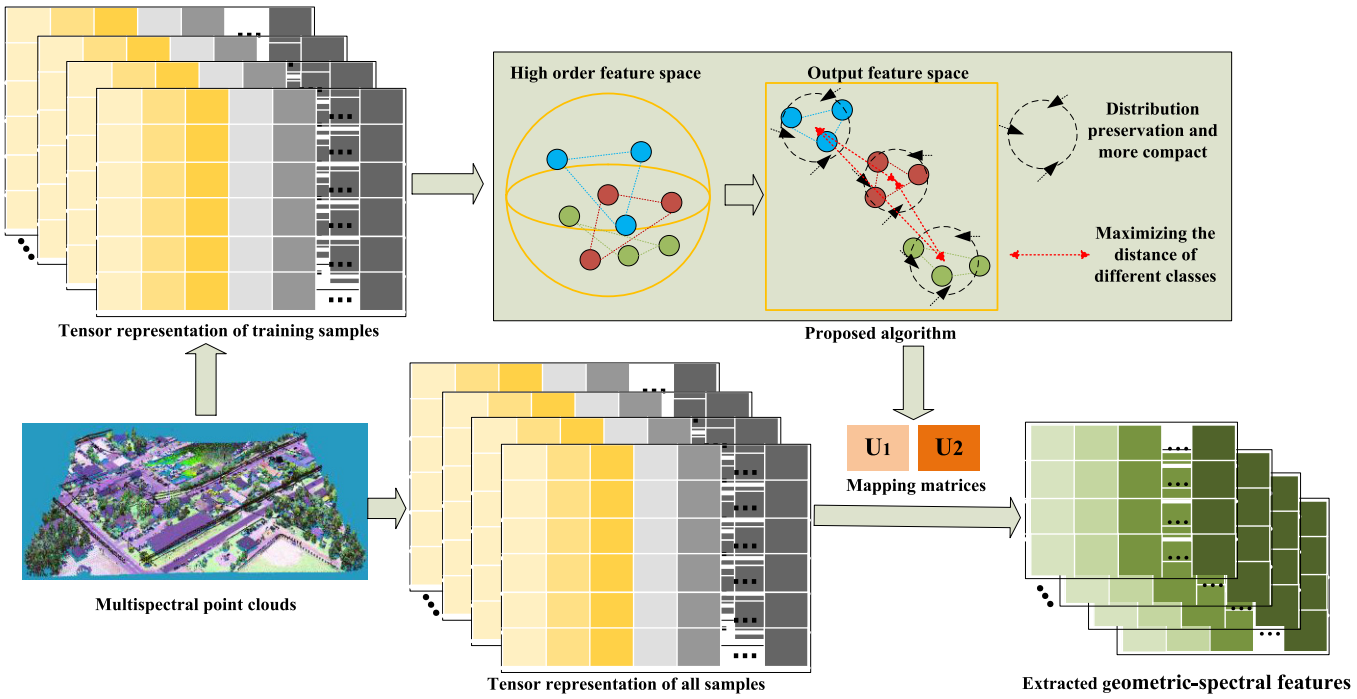


Fig. 3. Flowchart of the proposed method.

The spatial location and spectral information are connected in one mode, and the neighborhood geometric and spectral structures are connected in another mode.

III. TMDE Algorithm for Multispectral Point Clouds Feature Extraction

Given a training set $\mathcal{X} = \{\mathcal{X}_i\}, (i = 1, 2, \ldots, N)$ in high-dimensional space $\mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$, for any sample $\mathcal{X}_i$, the corresponding class label is $L_i \in \{1, 2, \ldots, C\}$, where C is the number of classes. The number of samples from class c is denoted by n_c, then $\sum_{c=1}^{C} n_c = N$. The purpose is to search for N mapping matrices $\{\mathbf{U}_i \in \mathbb{R}^{P_i \times I_i}, P_i \leq I_i, i = 1, 2, \ldots, M\}$ that map $\mathcal{X}_i \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ into $\mathcal{Y}_i \in \mathbb{R}^{P_1 \times P_2 \times \cdots \times P_M}$. That is,

$$\mathcal{Y}_i = \mathcal{X}_i \times_1 \mathbf{U}_1 \times \mathbf{U}_2 \times \cdots \times_M \mathbf{U}_M, \quad i = 1, 2, \ldots, N. \tag{6}$$

The TMDE algorithm aims to extract discriminative geometric–spectral features by preserving the intraclass samples' distribution and maximizing the distance of different classes, as shown in Fig. 3.

A. Intraclass Feature Compaction

We first construct an intraclass neighborhood graph:

$$\mathbf{G}_a = \begin{cases} e^{-(\|\mathcal{X}_i - \mathcal{X}_j\|^2/(2\sigma^2))}, & \text{if } \mathcal{X}_i \text{ is anong } k_1\text{NN of } \mathcal{X}_j \\ & \text{and } L_i = L_j \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where $\mathcal{X}_i$ and $\mathcal{X}_j$ come from the same class, k_1NN denotes the k_1 nearest neighbors (NN), σ is a scale parameter, which controls the smoothness of similarity measure, and σ is set to 1 in this article.

The feature distribution is the essential character of one class. Making feature distribution more compact can reduce the intraclass scatter, which enhances the identifiability of the class. Therefore, in the output-reduced feature space, TMDE suggests that distances of samples from the same class are as small as possible and keeps the intraclass feature distribution to enhance the class separability. Based on this point, we apply the following optimization:

$$\arg \min_{\mathcal{Y}_1, \ldots, \mathcal{Y}_N} \left\{ \sum_{i,j}^{N} \mathbf{G}_a(i, j) \cdot \|\mathcal{Y}_i - \mathcal{Y}_j\|_F^2 \right\} \tag{8}$$

where $\mathbf{G}_a$ represents the local structure of the features from one class. According to (7), $\mathbf{G}_a(i, j)$ will get larger as the distance between $\mathcal{X}_i$ and $\mathcal{X}_j$ getting closer in the feature distribution. $\|\mathcal{Y}_i - \mathcal{Y}_j\|_F^2$ has a larger weight in the optimization problem (8), which makes $\|\mathcal{Y}_i - \mathcal{Y}_j\|_F^2$ be smaller. The smaller value of $\|\mathcal{Y}_i - \mathcal{Y}_j\|_F^2$ means that $\mathcal{Y}_i$ and $\mathcal{Y}_j$ are closer.

B. Interclass Discriminative Information Maximization

We also construct an interclass neighborhood graph

$$\mathbf{G}_b = \begin{cases} 1 - e^{-(\|\mathcal{X}_i - \mathcal{X}_j\|_F/(2\sigma^2))}, & \text{if } \mathcal{X}_i \text{ is among } k_2\text{NN of } \mathcal{X}_j \\ & \text{and } L_i \neq L_j \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $\mathcal{X}_i$ and $\mathcal{X}_j$ come from different classes. It is worth noting that larger weights are assigned to sample pairs with shorter distances, which helps to reduce aliasing between different classes.

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We now dissociate the adjacent samples from different classes from each other

$$\arg \max_{\mathcal{Y}_1,\ldots,\mathcal{Y}_N} \left\{ \sum_{i,j}^{N} \mathbf{G}_b(i,j) \cdot \|\mathcal{Y}_i - \mathcal{Y}_j\|_F^2 \right\}. \tag{10}$$

C. Objective Function and Solution

The final objective function is obtained by combining the above two objective functions based on the maximum margin criterion (MMC) [47] as follows:

$$\arg \min_{\mathcal{Y}_1,\ldots,\mathcal{Y}_N} \times \left\{ \sum_{i,j}^{N} \mathbf{G}_a(i,j) \cdot \|\mathcal{Y}_i - \mathcal{Y}_j\|_F^2 - \sum_{i,j}^{N} \mathbf{G}_b(i,j) \cdot \|\mathcal{Y}_i - \mathcal{Y}_j\|_F^2 \right\}. \tag{11}$$

MMC is a scatter-difference-based rule. It gives a good representation for class separability. Geometrically, it maximizes the average margin between classes. By using (6), (11) can be written as follows:

$$\begin{aligned} &\arg \min_{\{\mathbf{U}_m\}_{m-1}^{M}} \times \left\{ \begin{array}{l} \sum_{i,j}^{N} \mathbf{G}_a(i,j) \cdot \|(\mathcal{X}_i - \mathcal{X}_j) \times_1 \mathbf{U}_1 \times \mathbf{U}_2 \times \ldots \times_M \mathbf{U}_M\|_F^2 \\ -\sum_{i,j}^{N} \mathbf{G}_b(i,j) \cdot \|(\mathcal{X}_i - \mathcal{X}_j) \times_1 \mathbf{U}_1 \times \mathbf{U}_2 \times \ldots \times_M \mathbf{U}_M\|_F^2 \end{array} \right\} \\ &= \arg \min_{\{\mathbf{U}_m\}_{m-1}^{M}} \times \left\{ \sum_{i,j}^{N} (\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \left\| (\mathcal{X}_i - \mathcal{X}_j) \prod_{m=1}^{M} \times_m \mathbf{U}_m \right\|_F^2 \right\}. \end{aligned} \tag{12}$$

The objective function (12) is a multiparameter and nonconvex problem. The mapping matrices $\{\mathbf{U}_m\}_{m=1}^{M}$ cannot be obtained once by directly solving (12). Inspired by the alternating optimization, we obtain $\{\mathbf{U}_m\}_{m=1}^{M}$ one by one through an iterative process. Here, we assume that only the mapping matrix $\mathbf{U}_m$ is unknown and we iteratively optimize $\mathbf{U}_m$ from $m = 1$ to $m = M$ by fixing other $M-1$ mapping matrices $\mathbf{U}_1, \ldots, \mathbf{U}_{m-1}, \mathbf{U}_{m+1}, \ldots, \mathbf{U}_M$. To facilitate description, when the $M-1$ times of n-mode products are conducted on all modes except the mth mode, i.e., $\mathcal{X} \times_1 \mathbf{U}_1 \times_2 \cdots \times_{m-1} \mathbf{U}_{m-1} \times_{m+1} \mathbf{U}_{m+1} \times_{m+2} \cdots \times_M \mathbf{U}_M$, we denote this procedure as $\mathcal{X} \prod_{j \neq m}^{M} \times_j \mathbf{U}_j$. For an arbitrary sample $\mathcal{X}_i$, we denote $\mathcal{Z}_i = \mathcal{X}_i \prod_{j \neq m}^{M} \times_j \mathbf{U}_j$ and then its low-dimensional representation is $\mathcal{Y}_i = \mathcal{Z}_i \times_m \mathbf{U}_m$. Therefore, (12) reduces to

$$\arg \min_{\mathbf{U}_m} \left\{ \sum_{i,j}^{N} (\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \|(\mathcal{Z}_i - \mathcal{Z}_j) \times_m \mathbf{U}_m\|_F^2 \right\}. \tag{13}$$

Based on the tensor property $\|\mathcal{H}\| = \|\mathbf{H}^{(k)}\|_F$ where $\mathbf{H}^{(k)}$ is the k-mode unfolding of tensor $\mathcal{H}$, we have $\|\mathcal{Z} \times_m \mathbf{U}_m\|_F = \|\mathbf{U}_m \mathbf{Z}^{(m)}\|_F$. We can further rewrite (13) as

$$\begin{aligned} &\arg \min_{\mathbf{U}_m} \left\{ \sum_{i,j}^{N} (\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \|(\mathcal{Z}_i - \mathcal{Z}_j) \times_m \mathbf{U}_m\|_F^2 \right\} \\ &= \arg \min_{\mathbf{U}_m} \left\{ \sum_{i,j}^{N} (\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \left\| \mathbf{U}_m \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right) \right\|_F^2 \right\}. \end{aligned} \tag{14}$$

Moreover, by using $\|\mathbf{H}\|_F^2 = \mathrm{Tr}(\mathbf{H}\mathbf{H}^T)$, we have

$$\begin{aligned} &\arg \min_{\mathbf{U}_m} \left\{ \sum_{i,j}^{N} \left(\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j) \right) \cdot \left\| \mathbf{U}_m \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right) \right\|_F^2 \right\} \\ &= \arg \min_{\mathbf{U}_m} \left\{ \sum_{i,j}^{N} (\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \mathrm{Tr}\left[\mathbf{U}_m \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right) \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right)^T \mathbf{U}_m^T \right] \right\} \\ &= \arg \min_{\mathbf{U}_m} \left\{ \sum_{i,j}^{N} \mathrm{Tr}\left[(\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \mathbf{U}_m \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right) \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right)^T \mathbf{U}_m^T \right] \right\} \\ &= \arg \min_{\mathbf{U}_m} \left\{ \mathrm{Tr} \left\{ \mathbf{U}_m \sum_{i,j}^{N} \left[(\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right) \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right)^T \right] \mathbf{U}_m^T \right\} \right\}. \end{aligned} \tag{15}$$

To avoid trivial solutions, the constraint $\mathbf{U}_m \mathbf{U}_m^T = \mathbf{I}$, $(m = 1, \ldots, M)$ is imposed to uniquely determine the mapping matrices. Define

$$\mathbf{F}_m = \sum_{i,j}^{N} \left[\left(\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j) \right) \cdot \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right) \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)} \right)^T \right]. \tag{16}$$

Then, we further reduce the optimization of $\mathbf{U}_m$ to

$$\arg \min_{\mathbf{U}_m \mathbf{U}_m^T = \mathbf{I}} \left\{ \mathrm{Tr}\left\{ \mathbf{U}_m \mathbf{F}_m \mathbf{U}_m^T \right\} \right\}. \tag{17}$$

The problem (17) can be solved by the eigenvalue decomposition of $\mathbf{F}_m$ as follows [48]:

$$\mathbf{F}_m \mathbf{E}_m = \mathbf{E}_m \mathbf{\Lambda}_m \tag{18}$$

where $\mathbf{E}_m \in \mathbb{R}^{I_m \times I_m}$ is constituted by eigenvectors. The solution $\mathbf{U}_m \in \mathbb{R}^{P_m \times I_m}$ of the minimization problem (17) is given by P_m eigenvectors associated with the least P_m eigenvalues.

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Algorithm 1 TMDE

Inputs: Training samples $\{\mathcal{X}_i\}_{i=1}^N$ and their class labels $\{L_i\}_{i=1}^N$. The retained dimensionality $P_1, \ldots, P_M$. The maximum number T of iterations and threshold parameter ε.

1: Construct the two neighborhood graphs $\mathbf{G}_a$ and $\mathbf{G}_b$ by (7) and (9), respectively.
2: Initialize mapping matrices $\{\mathbf{U}_m\}_{m=1}^M$ with identity matrices $\mathbf{U}_1^0 = \mathbf{I}_{I_1}, \ldots, \mathbf{U}_M^0 = \mathbf{I}_{I_M}$.
3: **for** $t = 1 : T$ **do**
4: **for** $m = 1 : \mathrm{M}$ **do**
5: Compute

$$\left\{\mathcal{Z}_i = \mathcal{X}_i \times_1 \mathbf{U}_1^{t-1} \times_2 \cdots \times_{m-1} \mathbf{U}_{m-1}^{t-1} \times_{m+1} \mathbf{U}_{m+1}^{m+1} \times_{m+2} \cdots \times_M \mathbf{U}_M^{t-1}\right\}_{i=1}^N,$$

and perform the mode-m flattening of $\mathcal{Z}_i$ to matrices: $\mathcal{Z}_i \Rightarrow_m \mathbf{Z}_i^{(m)}$.
6: Compute

$$\mathbf{F}_m^t = \sum_{i,j}^N \left[(\mathbf{G}_a(i,j) - \mathbf{G}_b(i,j)) \cdot \left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)}\right)\left(\mathbf{Z}_i^{(m)} - \mathbf{Z}_j^{(m)}\right)^T\right]$$

7: Perform eigenvalue decomposition on $\mathbf{F}_m^t$ and select P_k eigenvectors associated with the least P_k eigenvalues to form the mapping matrix $\mathbf{U}_m \in \mathbb{R}^{P_m \times I_m}$.
8: **end for**
9: **If** $t \geq 2$ and $\sum_{k=1}^M \|\mathbf{U}_k^t - \mathbf{U}_k^{t-1}\|_F < \varepsilon$, **break**
10: **end for**

Outputs: Set of mapping matrices $\mathbf{U}_m \in R^{P_m \times I_m}, (i = 1, 2, \ldots, M)$

It has been theoretically demonstrated that the alternating optimization procedure will converge to a local optimum based on the result that a monotone, bounded sequence of real numbers [49]. The aforementioned procedures for optimization are summarized in Algorithm 1.

After we obtain the final mapping matrices $\{\mathbf{U}_m^*\}_{m=1}^M$, all the samples can be mapped into the low-dimensional features space by mode-k product as $\mathcal{Y}_i = \mathcal{X}_i \times_1 \mathbf{U}_1^* \times \mathbf{U}_2^* \times \cdots \times_M \mathbf{U}_M^*$. In practice, for our multispectral point cloud data, two mapping matrices $\mathbf{U}_1^*, \mathbf{U}_2^*$ are obtained to extract the geometric–spectral features, which enhance the local discriminative information

$$\mathcal{Y}_i = \mathcal{X}_i \times_1 \mathbf{U}_1^* \times \mathbf{U}_2^* \tag{19}$$

where $\mathbf{U}_1^* \in \mathbb{R}^{P_1 \times I_1}, \mathbf{U}_2^* \in \mathbb{R}^{P_2 \times I_2}$. $\mathbf{U}_1^*$ reduces the dimensionality from I_1 to P_1 and integrates the location and spectral information of one multispectral point. $\mathbf{U}_2^*$ aims to reduce the dimensionality of the neighborhood features to P_2.

D. Computational Complexity

The major computational complexity of the proposed method is dominated by Steps 5, 6, and 7. For our multispectral point cloud data, the size of the input samples $\{\mathcal{X}_i\}_{i=1}^N$ is $I_1 \times I_2$, and the size of output features is $P_1 \times P_2$. For each iteration, the total computational complexity is the sum of the complexity of each mode. Specifically, for the case with the mode $m = 1$, the complexity of Steps 5, 6, and 7 is $O(NI_1I_2P_2)$, $O[C(k_1+k_2)(I_1^2P_2)]$, and $O(I_1^3)$, respectively. For the case with the mode $m = 2$, the complexity of Steps 5, 6, and 7 is $O(NI_1I_2P_1)$, $O[C(k_1+k_2)(I_2^2P_1)]$, and $O(I_2^3)$, respectively. If the proposed TMDE algorithm converges with t iterations, the upper bound of the total computational complexity is about $O\{t[NI_1I_2(P_1+P_2) + C(k_1+k_2)(I_1^2P_2 + I_2^2P_1) + I_1^3 + I_2^3]\}$. Although this is large when the dimension of each mode is very high, it can be done offline and the computational cost is acceptable.

IV. Data and Experimental Methods

A. Data Set Description

In our experiments, real multispectral point cloud data are used. The used multispectral point cloud data were collected by Optech Titan MS-LiDAR System that is the first operational multispectral airborne laser scanner available to the industry and scientific community. It has three independent active imaging channels that support 1550-, 1064-, and 532-nm wavelengths for multispectral mapping of the scene. Combining these three wavelengths can enhance the spectral measurement of the scene. The three wavelengths' lasers provide near-identical swaths from the three channels through a single scanning mechanism. The 532-nm channel is pointed at 7° forward of the nadir to minimize returns from the water surface and maximize the probability of water penetration. The 1064-nm channel is pointed to the nadir to maximize correlation with legacy LiDAR data sets collected with a 1064-nm wavelength also pointing to nadir. The 1550-nm channel is pointed at 3.5° forward of the nadir. Other specifications of the Titan system and data sets are given in Table I. Readers are recommended to read [21] and spec-sheet of Titan for more detailed information [50].

The data consist of three separate point clouds, each of which corresponds to a laser wavelength. Each return contains an intensity of a single laser wavelength. In this article, the multispectral point cloud refers to each point with multiple intensities corresponding to different channels. Therefore, we first synthetically created our multispectral point cloud, where each point contains its own detected intensity and two other assigned intensities from the other channels. The above step is similar to that described in [30]. We used the point cloud of 1550-nm channel as the benchmark and assign the intensity values of the other two channels for each point through the NN search in 3-D space. The reason for choosing the 1550-nm channel as the benchmark is that 1550-nm wavelength has the weakest penetration to water, and a 1550-nm channel usually has the fewest number of points. In practical work, we first searched for the nearest point in 1064- and 532-nm channels for each point in 1550-nm channel and then assigned the intensity values of the nearest points to the point in 1550-nm channel. Finally, we got

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TABLE I

SPECIFICATIONS OF THE SENSING CHANNELS OF THE MS-LIDAR DATA ACQUISITION SYSTEM AND THE MAIN CHARACTERISTICS OF THE EXPERIMENTAL DATA SETS

Parameter	Channel 1	Channel 2	Channel 3
Laser wavelength	1550 nm	1064 nm	532 nm
Beam divergence	$\approx$ 0.36 mRad	$\approx$ 0.3 mRad	$\approx$ 1 mRad
Look angle from Nadir	3.5 degrees forward	0 degrees	7.0 degrees forward
Pulse repletion frequency	50-300 kHz	50-300 kHz	50-300 kHz
Intensity quantization level	12 bit	12 bit	12 bit
Point density	Data set 1: 19.8 pts/m^2 Data set 2: 12.4 pts/m^2	Data set 1: 20.8 pts/m^2 Data set 2: 13.4 pts/m^2	Data set 1: 21.2 pts/m^2 Data set 2: 13.2 pts/m^2
Ground pulse footprint size	Data set 1: $\approx$ 0.18m Data set 2: $\approx$ 0.18m	Data set 1: $\approx$ 0.15m Data set 2: $\approx$ 0.15m	Data set 1: $\approx$ 0.5m Data set 2: $\approx$ 0.5m
Scene size	Data set 1: 600 m $\times$ 600 m; Data set 2: 595 m $\times$ 600 m		
Mean flying height	Data set 1: 500 m; Data set 2: 500 m		

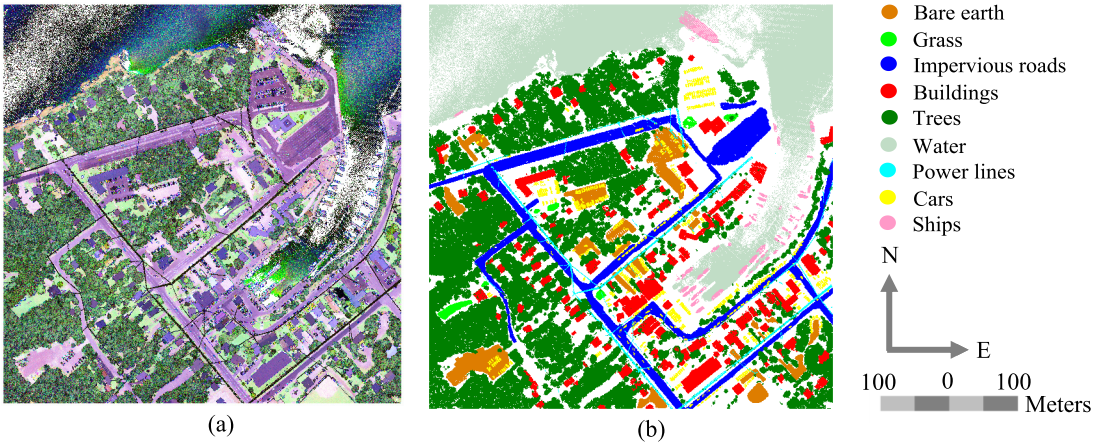


Fig. 4. Harbor of Tobermory data set. (a) Multispectral point clouds. (b) Ground truth.

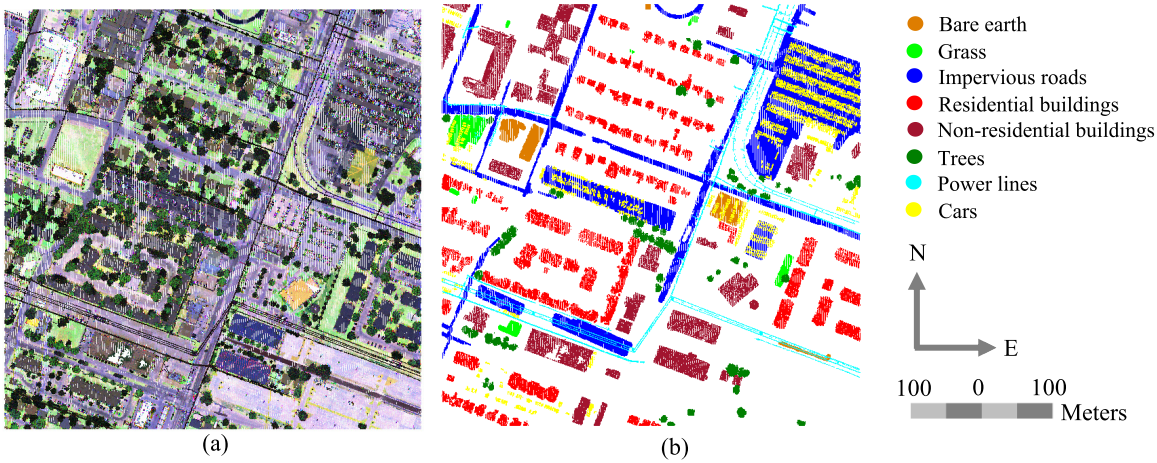


Fig. 5. University of Houston data set. (a) Multispectral point clouds. (b) Ground truth.

our multispectral point cloud data with three-wavelength intensity values: $[x, y, z, \mathrm{LRI}_{\lambda_{1550}}, \mathrm{LRI}_{\lambda_{1064}}, \mathrm{LRI}_{\lambda_{532}}]$. Experimental data were collected from two study areas, forming two data sets.

The first data set is the Harbor of Tobermory, where the scene covers a natural coastal area, located in Tobermory, ON, Canada. We classified the study areas as different classes according to heights, materials, and semantic information of land covers. The multispectral point clouds and ground truth are shown in Fig. 4. The land cover classes in the Harbor of Tobermory data set are bare earth, grass, impervious roads, buildings, trees, water, power lines, cars, and ships, which were identified by visual inspection of the MS-LiDAR image as well as 3-D visualization of point clouds and the RGB image on Google Earth.

The second data set is from a part of the University of Houston campus, Houston, TX, USA. Fig. 5 shows the multispectral point clouds and the ground truth, where eight

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classes of land covers are considered, consisting of bare earth, grass, impervious roads, residential buildings, nonresidential buildings, trees, power lines, and cars.

Due to the difference between the incident angle of the laser pulse and the orientation of the object surface, the intensity of a few points is very high, which makes the whole scene very dark. The low contrast is not conducive to classification. Histogram equalization was used to enhance the contrast of point cloud spectra.

Height is a very important discriminating feature for the classification task. We calculated the height above ground (HAG) to replace the height z of the original point clouds. The HAG of point i is calculated as $\mathrm{HAG}_i = z_i - \mathrm{GE}(x_i, y_i)$, where z_i is the elevation of point i and $\mathrm{GE}(x_i, y_i)$ is the ground elevation at the location of point i. We used the DSM function of ENVI LiDAR software with the input of a combined point cloud from all channels to generate the $\mathrm{GE}(x_i, y_i)$.

B. Experimental Setup

To evaluate the effectiveness of the proposed method, the features $[x, y, \mathrm{HAG}, \mathrm{LRI}_{\lambda_{1550}}, \mathrm{LRI}_{\lambda_{1064}}, \mathrm{LRI}_{\lambda_{532}}]$ from each single multispectral point are used as a benchmark, represented as No-neighbors. We directly vectorize the tensor representing each multispectral point into a feature vector as another contrast method, represented as Vectorization. We further compare the classification results of the proposed method with several representative methods (i.e., PCA [51], MPCA [42], TDLA [43], locality preserving projections (LPP) [52], and DGTDA [39]) in computer vision. The 95% and 90% of the cumulative variance was set by cross-validation (CV) to preserve features for PCA and MPCA, respectively. For LPP, the number of neighbors was set to 12, and the bandwidth parameter of the Gaussian kernel was set to 1 by default. The 100 features from LPP were preserved for classification. For TDLA, as suggested in [43], the number of NNs from the same class and the number of NNs from different classes were set to 6, and the scaling factor was set to 2. The feature dimensionality preserved by TDLA and DGTDA was set to the same as the proposed TMDE for a fair comparison. The classification was implemented by SVM classifier. The penalization parameter C of the SVM model was optimized by CV in the interval $C \in [10, 10^2, 10^3, \ldots, 10^7]$, and the slack variable $\xi = 10^{-7}$ was also selected by CV. These parameters may not be optimal but are considered reasonable after a study of a wide range of parameters. All the programs were run on the MATLAB platform. The CPU is Intel Core i7-7820X with a frequency at 3.6 GHz, and the RAM is 128 GB.

In each experiment, 200 training samples and 20 000 test samples for each class were randomly selected from the labeled point clouds on the Harbor of Tobermory data set. The 0.5% of the labeled samples were randomly selected as training samples and the rest as test samples on the University of Houston data set. Accuracy per class, classification confusion matrix, OA, average accuracy (AA), kappa statistics, McNemar's test [53], computation time, and overall classification maps are used to evaluate all methods.

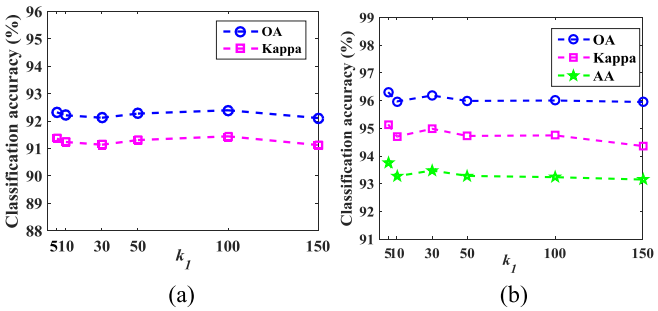


Fig. 6. Classification accuracy (%) versus k_1. (a) Harbor of Tobermory data set. (b) University of Houston data set.

C. Parameter Setup and Discussion

There are three kinds of important parameters that need to be determined for the TMDE algorithm.

1) Parameters for Constructing Neighborhood Graphs: The numbers of the nearest points k_1 and k_2 used for constructing neighborhood graphs were chosen as $k_2 = C \cdot k_1$ for simplicity. The classification accuracy with a varying k_1 is shown in Fig. 6. The changing range of classification accuracy with varying k_1 is very small, less than 0.3% for OA and less than 0.35% for kappa coefficient on the Harbor of Tobermory data set, and less than 0.6% for AA, less than 0.5% for kappa coefficient, and 0.32% for OA on the University of Houston data set, respectively. It is obvious that the numbers of the nearest points from one class and different classes have a little effect on the classification accuracy. Therefore, any integer from 5 to 150 can be adopted to set k_1. In our subsequent experiments, k_1 is set to 100 for the two data sets.

2) Size of Extracted Feature: Here, the number k of nearest points used for tensor representation is set as 100. In the experiment, the input tensor $\mathcal{X}_i \in \mathbb{R}^{100\times 6}$ is reduced to an output tensor of size $P_1 \times P_2$. In the mapped space, the feature dimensionality is P_1 and P_2, where P_1 corresponds to the intrinsic dimensionality of neighborhood space and P_2 corresponds to the intrinsic feature dimensionality of one multispectral point. Since the tensor is employed to describe the center point, the P_1 should be one intuitively. Numerically, P_1 and P_2 are related to the eigenvalues of F_1 and F_2, respectively. There are 100 eigenvalues for the first order and six eigenvalues for the second order since the size of the input tensor is 100×6. We experimentally show the distribution of eigenvalues sorted in the incrementing order in the first and second orders as Fig. 7. It can be seen that only one eigenvalue is negative and much smaller than the others that are nonnegative and very close to zero for the first order on both data sets. Therefore, the best choice of P_1 is 1 in terms of optimization. Note that, when $P_1 = 1$, the two-order tensor of size $P_1 \times P_2$ degrades to a vector of size P_2. For the second order, the six eigenvalues are all negative and the largest eigenvalue is -52.12. Thus, we have that $P_2 = 6$ on both data sets. The classification accuracies with the varying dimensionalities of P_1 and P_2 are presented in Fig. 8. The retained dimensionality P_1 and P_2 has a great effect on classification. The highest classification accuracy is achieved when $P_1 = 1$ and $P_2 = 6$ on both data sets, which also

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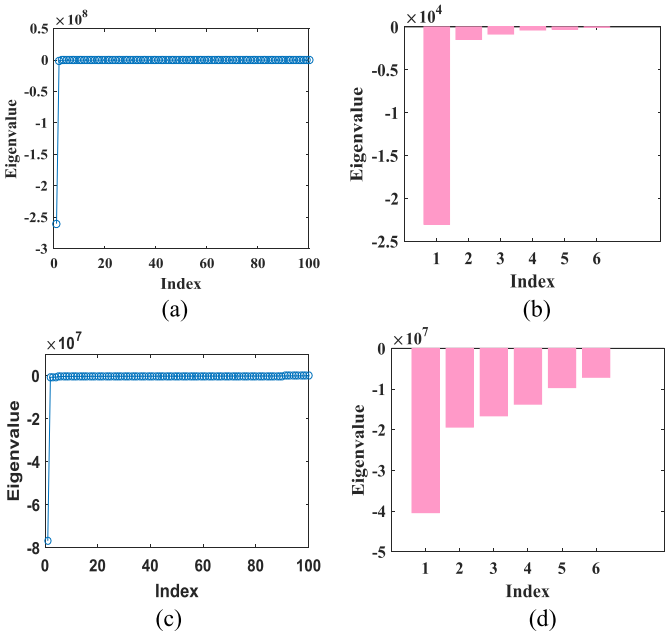


Fig. 7. Distribution of the eigenvalues in the first and second orders. (a) and (b) Harbor of Tobermory data set. (c) and (d) University of Houston data set. (a) and (c) First order. (b) and (d) Second order.

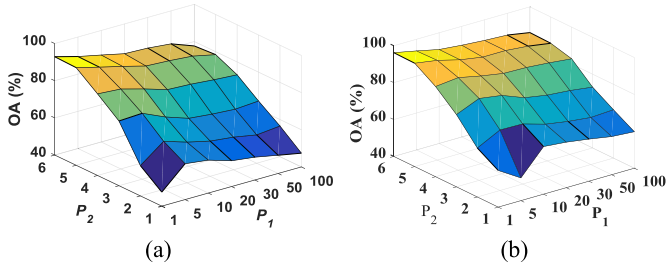


Fig. 8. OA (%) versus the retained dimensionality P_1 and P_2. (a) Harbor of Tobermory data set. (b) University of Houston data set.

suggests that the optimal values of P_1 and P_2 are 1 and 6, respectively. Therefore, the optimal size of P_i in our proposed TMDE algorithm can be set experimentally by analyzing the eigenvalues of matrix F_i in the ith mode optimization. In our subsequent experiments, P_1 and P_2 are set to 1 and 6 on both data sets, respectively.

3) Parameters for Tensor Representation: The number k of nearest points determines the field of view for tensor representation. The parameter k plays a great role in classification performance since the geometric and stereo spectral information is contained in adjacent points. When k is in an interval from 1 to a certain integer value, 100 in our case, a larger number of k means that more adjacent points are used in tensor representation, which can give a better performance, as shown in Fig. 9. With the increase of k from 10 to 100, the OA and kappa coefficient were improved by 7.6% and 7.8% on the Harbor of Tobermory data set, and 6% and 6.3% on the University of Houston data set. In addition, AA was improved by 5.82% on the University of Houston data set, which demonstrates that the tensor representation plays a signification role in improving the classification performance

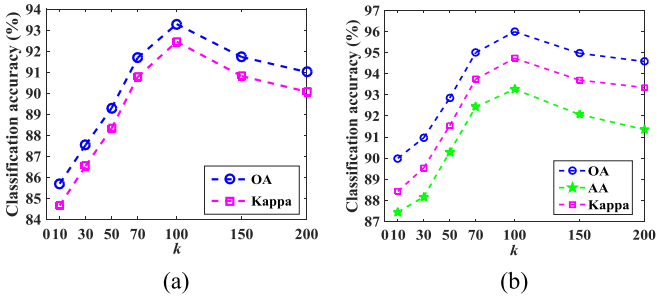


Fig. 9. Classification accuracy (%) versus k. (a) Harbor of Tobermory data set. (b) University of Houston data set.

of multispectral point clouds. When k is relatively large, the adjacent points may come from different classes, which will bring false geometric and spectral information. Therefore, excessive k will increase the intraclass dispersion and reduce the separability. In subsequent experiments, the number k of nearest points for tensor representation is set as 100.

V. Experimental Results

A. 3D Land Cover Classification

In the mapped feature space, six geometric–spectral features shown in Figs. 10 and 11 were extracted from multispectral point cloud data on both data sets for 3-D land cover classification. Tables II and III show the numerical classification results on the Harbor of Tobermory data set and the University of Houston data set, respectively. The classification results consist of the accuracy of each class, OA, kappa coefficient, and AA. As the results of No-neighbors shown, 83.19% of OA and 81.09% of kappa coefficient can be achieved on the Harbor of Tobermory data set when only single multispectral point features $[x, y, \mathrm{HAG}, \mathrm{LRI}_{\lambda_{1550}}, \mathrm{LRI}_{\lambda_{1064}}, \mathrm{LRI}_{\lambda_{532}}]$ were used. No-neighbors achieved 86.84% of OA, 82.70% of kappa coefficient, and 78.71% of AA on the University of Houston data set. The results of using the features of a single multispectral point can be used as a benchmark to evaluate the performance of neighborhood-based methods. Neighborhood-based methods can be compared to the benchmark method in terms of their accuracy statistics. Vectorization has a poor performance for classification. Vectorization vectorizes the 100×6 matrix representing a multispectral point into a vector with 600-D features that makes the classification to be a small sample problem. Among the remaining methods, 100-D features in LPP on both data sets, 144-D features in PCA on the Harbor of Tobermory data set and 106-D features in PCA on the University of Houston data set, were retained for classification. The input to the above methods is a feature vector. In the tensor-based methods, MPCA retained 6- $\times$ 4-D features that mean it retained 6-D features in the first order and 4-D features in the second order of the input tensor for classification. A total of 24-D features were finally used in MPCA for classification on both data sets. In the same way, TDLA, DGTDA and TMDE retained 1- $\times$ 6-D features for classification. TMDE is optimal in terms of classification accuracy. The 93.13% of OA and 92.27% of kappa coefficient

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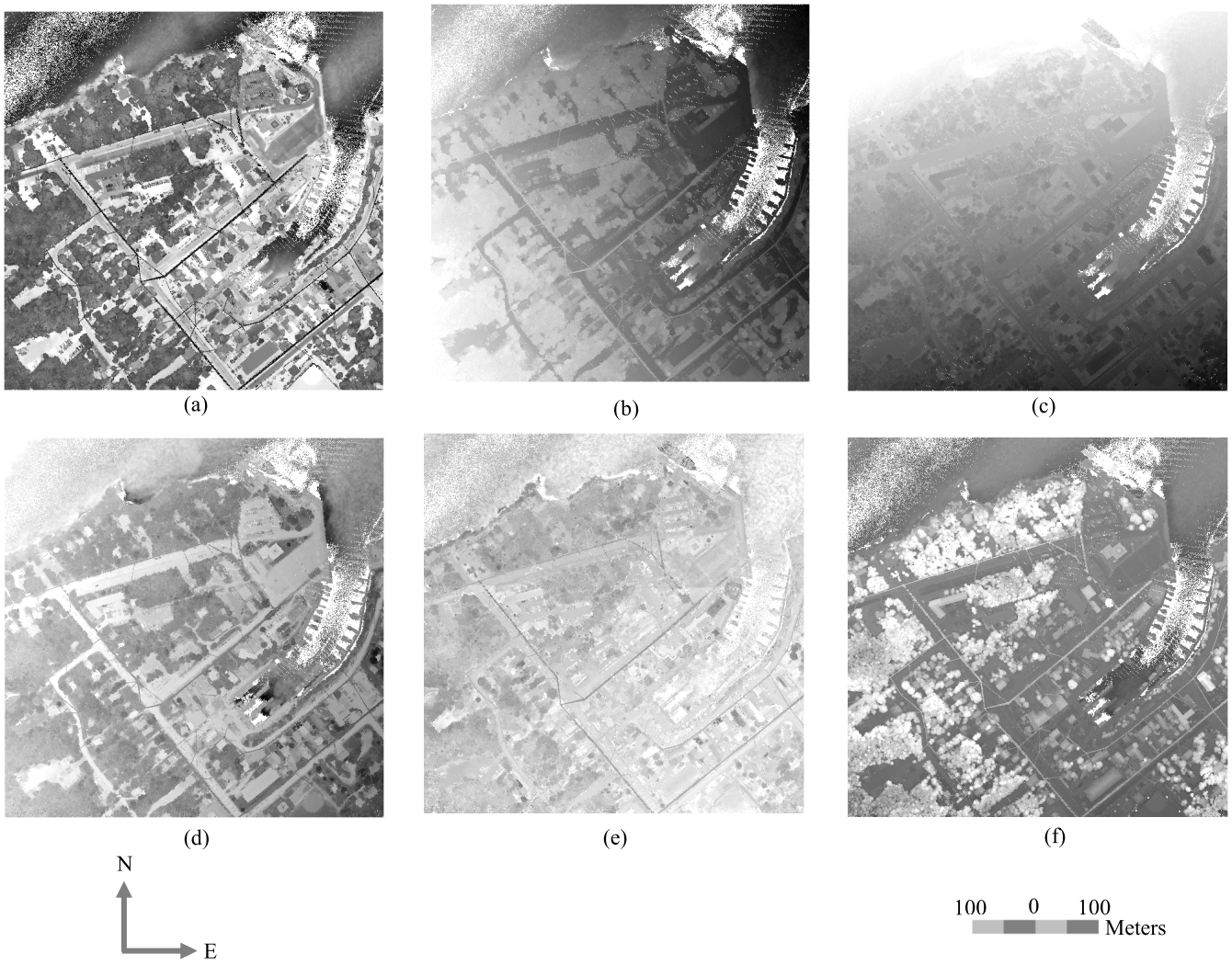


Fig. 10. Geometric–spectral features extracted from multispectral point cloud data for the Harbor of Tobermory data set when $P_1 = 1$ and $P_2 = 6$. (a)–(f) Corresponding eigenvalues increment sequentially.

were achieved by the proposed TMDE for 3-D land cover classification of nine classes on the Harbor of Tobermory data set. For the eight classes of the University of Houston data set, TMDE achieved 96.62% of OA, 95.56 of kappa coefficient, and 94.05 of AA. Compared with No-neighbors, the use of neighborhood points achieved an improvement with 9.94% of OA and 11.18% of kappa coefficient on the Harbor of Tobermory data set, 9.78% of OA, 12.86% of kappa coefficient, and 15.34% of AA on the University of Houston data set, respectively. Compared with the best contrast method, TMDE brought about an improvement with 2.45% of OA and 2.75% of kappa coefficient on the Harbor of Tobermory data set, and 3.71% of OA, 4.88% of kappa coefficient, and 3.51% of AA on the University of Houston data set.

The highest classification accuracy of each class is highlighted with bold text in Tables II and III. In the nine classes of the Harbor of Tobermory data set, TMDE achieved the highest accuracy for four classes. Furthermore, the highest classification accuracy of five classes in the eight classes of the University of Houston data set was achieved by TMDE. Classification confusion matrices of TMDE on the two data sets are given in Fig. 12. Among all the nine classes of land covers, the classification accuracy of power line is the worst. The power line point clouds were mainly misclassified into trees and buildings.

We conducted a McNemar's test between the proposed TMDE and each of the seven contrast methods (a total of seven tests) to further analyze the statistical difference between the proposed TMDE with others [53] Table IV shows the McNemar's test results. McNemar's test scores are interpreted as follows: when the score is 0, the two methods are considered to show similar performance. As this score diverges from 0 in the positive direction, it is indicated that their performance differs significantly [54]. By looking at the McNemar's test results, one can conclude that the proposed TMDE has performed significantly better results than contrast methods. Furthermore, all the McNemar's test scores are greater than 2.576, which indicates that TMDE is better than contrast methods with a confidence level of more than 99.5% on both data sets.

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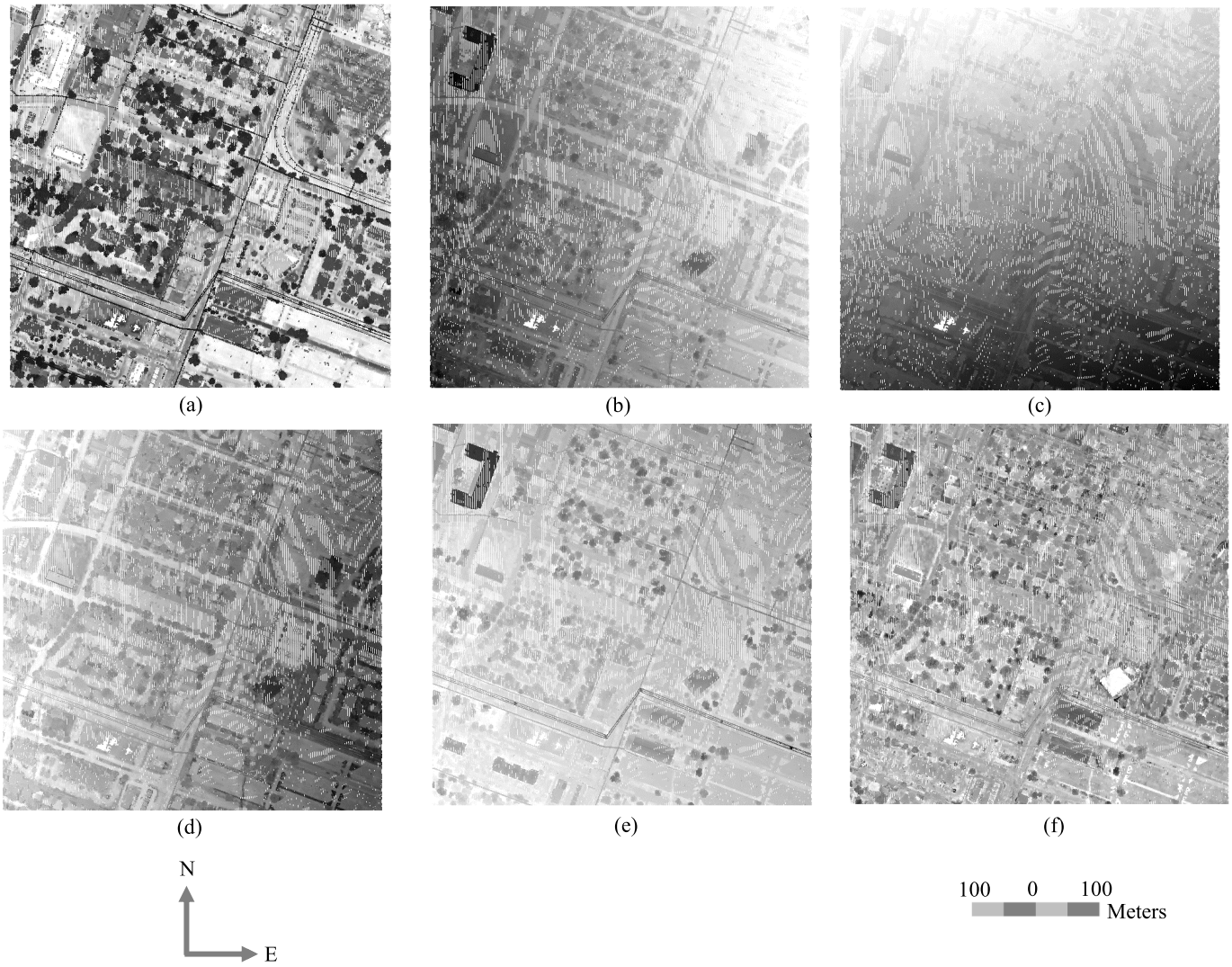


Fig. 11. Geometric–spectral features extracted from multispectral point cloud data for the University of Houston data set when $P_1 = 1$ and $P_2 = 6$. (a)–(f) Corresponding eigenvalues increment sequentially.

TABLE II
CLASSIFICATION ACCURACY (%) OF VARIOUS FEATURES ON THE HARBOR OF TOBERMORY DATA SET

Methods	Number of features	Bare earth	Grass	Impervious roads	Buildings	Trees	Water	Power lines	Cars	Ships	OA	Kappa
No-neighbours	6	90.56	92.92	84.07	89.39	81.30	93.50	90.22	75.85	50.93	83.19	81.09
Vectorization	600	83.03	90.25	74.51	58.36	62.03	14.72	48.31	99.62	16.04	60.76	55.86
LPP	100	91.93	97.14	90.05	88.54	82.35	95.52	90.33	75.03	65.72	86.29	84.57
PCA	144	91.32	96.30	90.19	88.83	81.23	94.98	88.01	76.17	64.14	85.68	83.89
MPCA	24	92.62	96.60	93.32	87.87	77.56	97.36	88.11	75.86	73.28	86.95	85.32
TDLA	6	93.38	97.79	95.18	**92.83**	87.24	98.46	**90.92**	81.09	72.66	89.95	88.69
DGTDA	6	**97.68**	98.51	**96.28**	89.43	77.07	98.41	85.34	83.96	**89.46**	90.68	89.52
TMDE	6	96.19	**99.78**	94.95	89.33	**90.15**	**98.99**	87.54	**93.08**	88.21	**93.13**	**92.27**

Table V reports the CPU time of each method. Although the computation time required by the proposed TMDE is less than TDLA, the computational speed is still not efficient enough. It can be improved by using the parallel implementation to reduce the running time, such as using CUDA in GPU and opening the parallel computing architecture in MATLAB to use the multicore processor in the computer.

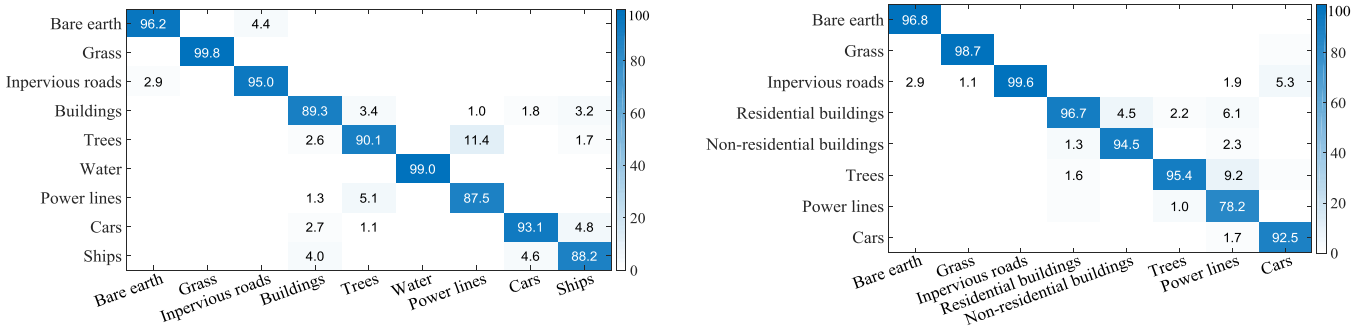
The thematic maps of different methods on the Harbor of Tobermory data set and University of Houston data set are shown in Figs. 13 and 14, respectively. Figs. 13(a) and 14(a) display color-coded point clouds classified by independently using each multispectral point. Notably, there is salt-and-pepper noise in the classification results. For example, some points of trees were incorrectly classified into power line

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TABLE III

CLASSIFICATION ACCURACY (%) OF VARIOUS FEATURES ON THE UNIVERSITY OF HOUSTON DATA SET

Methods	Number of features	Bare earth	Grass	Impervious roads	Residential buildings	Non-residential buildings	Trees	Power lines	Cars	OA	Kappa	AA
No-neighbours	6	58.70	84.56	98.46	83.52	68.29	80.05	63.82	92.30	86.84	82.70	78.71
Vectorization	600	78.83	74.53	**99.97**	84.27	63.71	73.38	47.71	1.55	78.12	68.64	65.49
LPP	100	83.68	89.18	97.61	87.32	77.07	86.24	74.32	85.81	89.64	86.39	85.15
PCA	106	80.27	86.94	97.43	86.41	74.44	84.65	72.41	86.73	88.76	85.25	83.66
MPCA	24	80.79	89.07	97.70	86.27	74.83	86.18	68.18	82.19	88.59	84.99	83.15
TDLA	6	95.08	97.54	94.64	**96.80**	93.68	94.87	**84.96**	66.73	92.24	89.78	90.54
DGTDA	6	94.05	92.38	98.19	93.56	89.03	91.12	75.97	80.29	92.91	90.68	89.32
TMDE	6	**96.82**	**98.65**	99.64	96.71	**94.54**	**95.38**	78.15	**92.50**	**96.62**	**95.56**	**94.05**



(a) (b)

Fig. 12. Classification confusion matrix of the proposed TMDE. The numerical values represent the percentage (%) of test samples of the corresponding class. (a) Harbor of Tobermory data set. (b) University of Houston data set.

TABLE IV

MCNEMAR'S TEST BETWEEN THE PROPOSED METHOD AND OTHERS

Data set	Classifier	No-neighbours	Vectorization	LPP	PCA	MPCA	TDLA	DGTDA
Harbor of Tobermory	TMDE	92.14	217.20	68.17	73.15	62.70	34.83	27.35
University of Houston	TMDE	242.88	365.61	193.17	213.17	216.66	115.95	116.05

TABLE V

CPU TIME (S) CONSUMED BY DIFFERENT METHODS ON THE HARBOR OF TOBERMORY DATA SET

LPP	PCA	MPCA	TDLA	DGTDA	TMDE
7161	11	15	36568	63	11379

and building. The reason for that is No-neighbors does not take the spatial neighborhood information into account in the classification process. Vectorization classified most points into the same class (cars for Harbor of Tobermory data set and impervious roads for University of Houston data set), as shown in Figs. 13(b) and 14(b). The reason for that is that directly vectorizing the matrix into a feature vector results in excessive feature dimensions and reduces the classification ability. Figs. 13(c)–(h) and 14(c)–(h) show the classification maps of LPP, PCA, MPCA, TDLA, DGTDA, and the proposed TMDE, respectively. These methods reduce the phenomenon of salt-and-pepper noise in classification maps and improve the ability of land cover classification by using neighborhood points. However, it does not mean that neighborhood points will always improve the classification performance. In the lower right corner of the Harbor of Tobermory data set, all the contrast methods using neighborhood points failed to classify grass, and only the proposed TMDE achieved the correct classification.

B. Classification of Fine Features

The classification maps of an area in the University of Houston data set were enlarged and shown in Fig. 15. In the classification map of TMDE, there is obvious homogeneity within the same class. The edges of different classes are clearly defined. It can be visually seen that TMDE can better classify point clouds, especially for trees and cars. The individual cars can be easily seen in the classification map of TMDE. It is worth pointing out that the cars under the trees were correctly classified by TMDE.

To evaluate the ability to classify fine features, we selected the training and test samples from heterogeneous areas where there are many classes in a small area. Fig. 16 shows the

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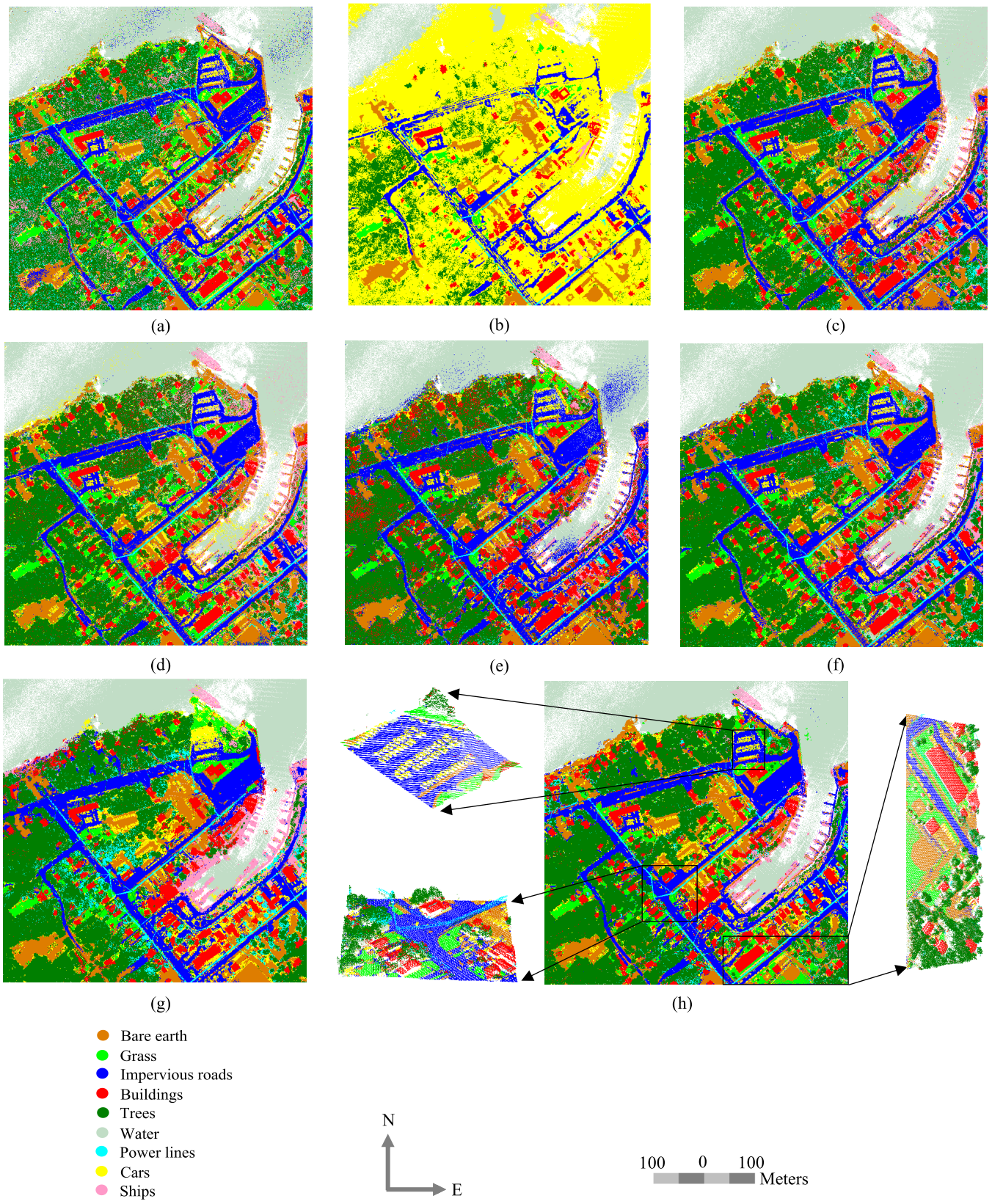


Fig. 13. Overall classification maps of the Harbor of Tobermory data set using different methods. (a) No-neighbors. (b) Vectorization. (c) LPP. (d) PCA. (e) MPCA. (f) TDLA. (g) DGTDA. (h) TMDE.

distribution of heterogeneous areas selected for the classification of fine features. Cars park on grasses and impervious roads in the selected heterogeneous areas. Trees grow to the roof of residential buildings. These conditions make two points that are very close are not in the same class. The neighborhood of a car point may contain a lot of grass or impervious

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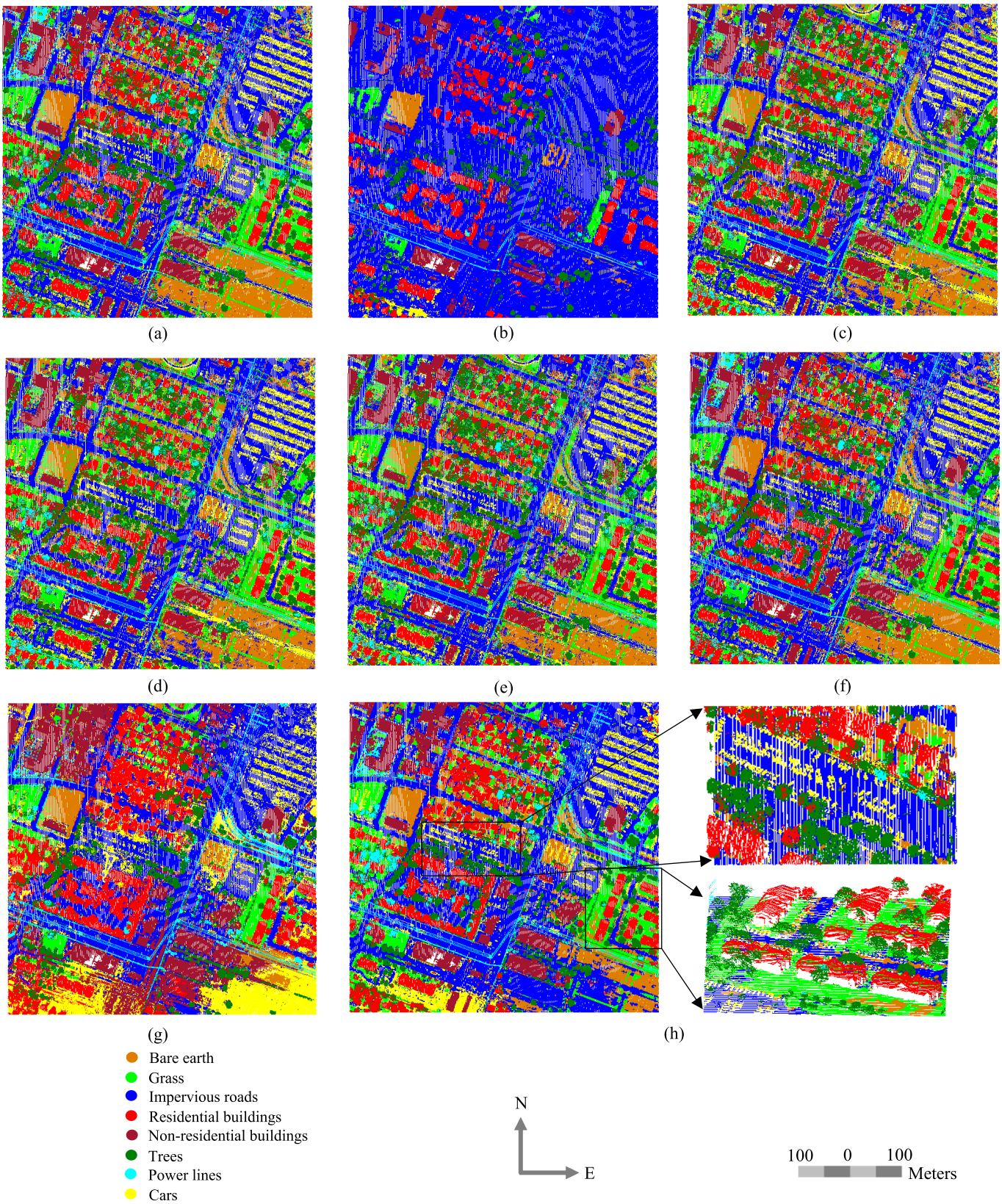


Fig. 14. Overall classification maps of the University of Houston data set using different methods. (a) No-neighbors. (b) Vectorization. (c) LPP. (d) PCA. (e) MPCA. (f) TDLA. (g) DGTDA. (h) TMDE.

road points. Similarly, the neighborhood of a tree point may contain a lot of residential building points. Such distribution of fine features increases the difficulty of classification. TMDE achieves the highest classification accuracy for fine features such as cars, grass, trees, and residential buildings. In the selected heterogeneous areas, the distribution of power lines

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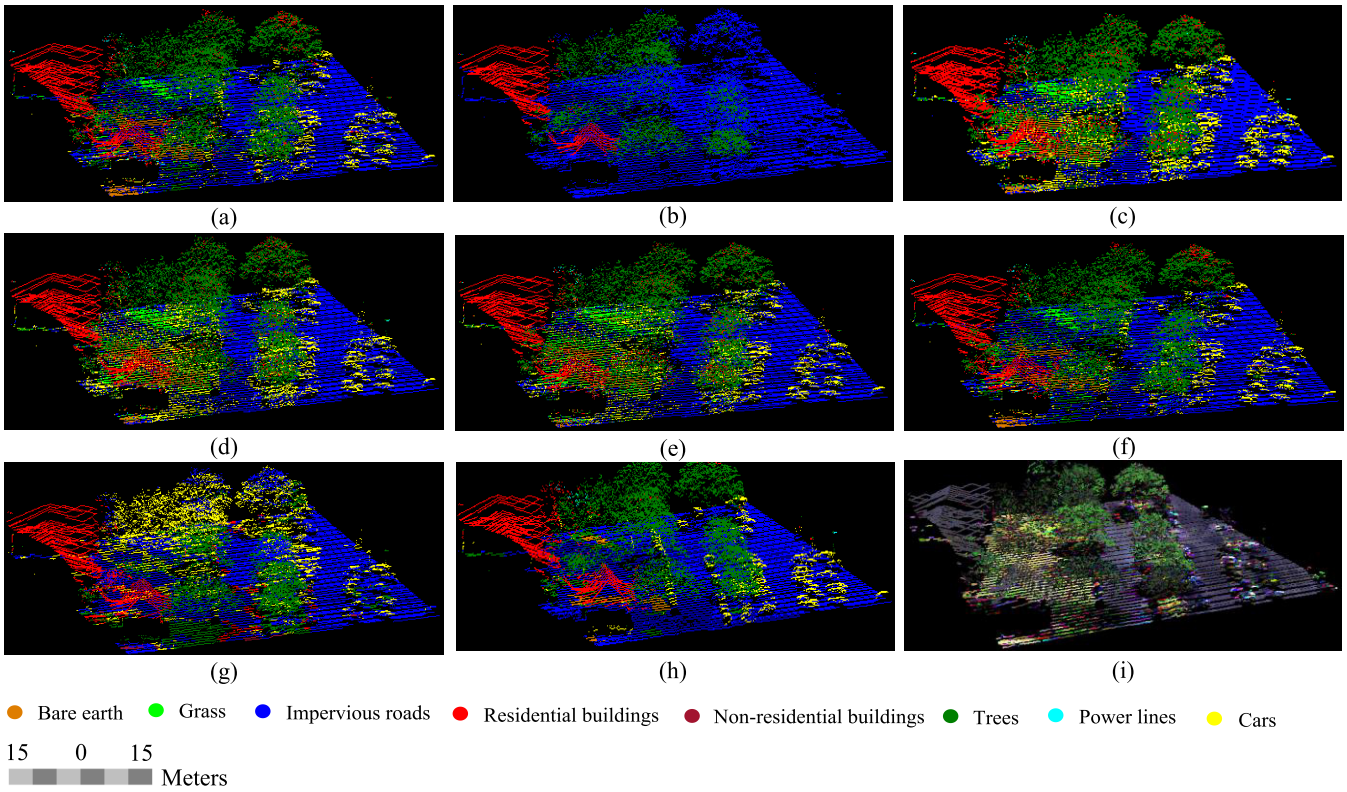


Fig. 15. 3-D view of part of the classified point cloud of the University of Houston data set. (a) No-neighbors. (b) Vectorization. (c) LPP. (d) PCA. (e) MPCA. (f) TDLA. (g) DGTDA. (h) TMDE. (i) Reference scene.

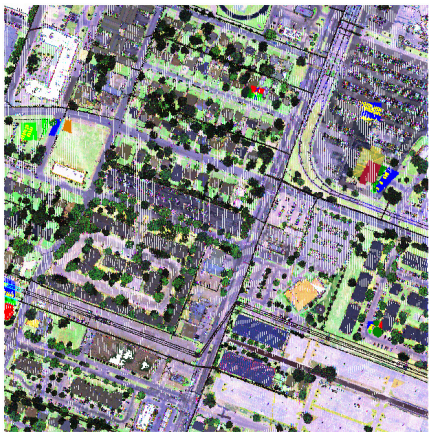


Fig. 16. Distribution of heterogeneous areas selected for the classification of fine features.

is relatively independent compared to the above-mentioned classes. There are no other classes in the neighborhood of the power line point. DGTDA did better for power line classification, but we believe that the proposed TMDE can be better when classifying the power line points under the situation that the trees grow close to power lines. Table VI shows the number of training and test samples. Table VII shows the numerical classification results. Among the eight classes, TMDE achieved the highest classification accuracy in five classes. Compared with the best contrast method, TMDE brought about an improvement with 3.49% of OA, 4.37% of kappa, and 3.77% of AA. This is due to the fact that we consider both making the distribution of the same class more compact and increasing the distance between different classes in the algorithm design.

C. Discussion

If each point is considered independently in classification, the phenomenon of salt-and-pepper noise usually appears in the classification results, as shown in Figs. 13(a) and 14(a). Incorporating spatial neighborhood information along with spectral information will alleviate the phenomenon of salt-and-pepper noise and increase classification accuracy greatly, as shown in Figs. 13(c)–(h) and 14(c)–(h). Among the neighborhood-based methods, LPP projects the data to a subspace, where the local neighborhood structure of the data is optimally preserved in a certain sense. LPP can represent the raw data with fewer features while preserving the local neighborhood structure. This makes the classification model easier to train and needs fewer training samples. However, it does not adjust the data distribution for classification. Therefore, there is no guarantee that the classification performance will be improved. PCA and MPCA project the data along with the directions of maximal variance and try to maximize the total scatter of data after a linear transformation. However, the data with a large intraclass scatter can also have a large total scatter even when it has a small interclass scatter. Therefore, the features extracted by PCA and MPCA cannot guarantee the improvement of classification, either. TDLA, DGTDA, and the proposed TMDE use different optimization ideas to project the data to another feature space, where the distribution of

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TABLE VI
NUMBER OF THE TRAINING AND TEST SAMPLES SELECTED FROM HETEROGENEOUS AREAS

Class	Bare earth	Grass	Impervious roads	Residential buildings	Non-residential buildings	Trees	Power lines	Cars	Total
Training	50	98	145	50	53	67	50	50	563
Test	1696	9740	14404	4000	5272	6676	549	3584	45921

TABLE VII
CLASSIFICATION ACCURACY (%) OF VARIOUS FEATURES ON THE HETEROGENEOUS AREAS IN THE UNIVERSITY OF HOUSTON DATA SET

Methods	Number of features	Bare earth	Grass	Impervious roads	Residential buildings	Non-residential buildings	Trees	Power lines	Cars	OA	Kappa	AA
No-neighbours	6	77.83	81.63	90.66	87.83	97.36	80.47	95.45	87.95	87.16	84.13	87.40
Vectorization	600	94.10	55.60	**99.12**	61.98	75.97	49.67	77.05	1.40	68.73	58.51	64.36
LPP	100	99.88	84.17	92.98	91.25	98.69	89.93	95.63	76.31	90.16	87.80	91.11
PCA	106	99.71	84.47	92.88	90.85	98.90	90.61	96.72	82.42	90.76	88.56	92.07
MPCA	24	**100.00**	69.36	94.72	82.93	98.50	91.15	95.08	49.19	84.88	81.06	85.12
TDLA	6	**100.00**	84.56	90.68	93.05	98.46	91.24	98.00	78.88	90.07	87.75	91.86
DGTDA	6	99.94	91.37	92.18	93.63	99.37	94.08	**98.18**	62.39	91.27	89.15	91.39
TMDE	6	99.59	**92.97**	93.56	**96.88**	**99.66**	**95.85**	96.17	**90.37**	**94.76**	**93.52**	**95.63**

samples from the same class will be more compact, and the distribution of samples from different classes will be more dispersed. This adjustment of the sample distribution is helpful for classification. TDLA measures the sample distribution by calculating the Euclidean distance between samples. DGTDA uses scatter matrices to measure the sample distribution. Neither of the two methods considers the relative relationship between the samples. They treat each sample equally in the optimization process. However, in actual situations, different samples will affect feature extraction to varying degrees. For example, samples distributed at the edge or samples distributed in aliasing regions will have a greater impact on feature extraction than the samples distributed at the center. Therefore, it is not optimal to treat each sample equally. Compared with TDLA and DGTDA, TMDE constructs an intraclass neighborhood graph and an interclass neighborhood graph to deal with each sample adaptively. According to the sample distribution, the weights of samples are adaptively assigned in optimization. Therefore, TMDE can extract better features for classification.

Nevertheless, neighborhood-based methods may over-smooth some boundaries or line-type targets, such as power line, due to the fact that the tensor characterizing the points of boundary or line targets will incorporate many points from other classes. All neighborhood-based methods face this problem and cannot eliminate it completely. This problem stems from the selection of neighborhood points. For a point on a boundary or linear target, it is impossible to select points with the same class to form the neighborhood by using a simple Euclidean distance metric. The more points in the neighborhood belong to the class of the point of interest, and the problem of over-smoothing the boundary or linear target can be better improved. A practical method is to screen the points when selecting the neighborhood points. The dimensionality features can be calculated on spherical neighborhoods at various radius sizes. The Shannon entropy criterion can be used to retrieve the optimal neighborhood size for each point [55]. One can use the intensity and dimensionality features to screen the neighborhood points so that the points that make up the neighborhood are in the same class as the point of interest, as far as possible. In practice, the difference of intensity and dimensionality features between the points in the neighborhood and the point of interest can be required to be less than a certain threshold value.

VI. CONCLUSION

In this article, we proposed a model based on tensor representation to classify MS-LiDAR point clouds for 3-D land cover mapping. To combine the spatial and spectral information, each point is represented as a second-order tensor by using the multispectral point and its neighborhood points. Then, we introduced the TMDE algorithm to explore the 3-D geometry and spectral structure information of multispectral point clouds in the low-dimensional feature space for the subsequent classification. The proposed TMDE algorithm has two main advantages: 1) the tensor representation takes advantage of the spatial and spectral information and 2) TMDE can preserve the intraclass samples' distribution, further make it more compact and maximize the distance of different classes to extract the discriminative features for classification. The experiments were carried out on two real multispectral point cloud data sets. The experimental results show that spatial neighborhood information has important contributions to classification. TMDE is more effective in the utilization of spatial and spectral information of multispectral point cloud, and furthermore, it has the ability to better classify fine features.

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